Technical specification template: Standby Generator Sets

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for standby generator sets for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- ISO 8528 series (reciprocating engine driven generating sets)
- IEC 60034 series (rotating electrical machines)
- Local emissions, noise and fuel storage regulations [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Generators per power path	3	PRELIMINARY_ASSUMPTION. N+1: the load is carried with the largest unit out
Rated apparent power (each)	2,000 kVA	UNVERIFIED_PLACEHOLDER. ASSUMED to be a PRP rating. NOT confirmed. ISO 8528: PRP, ESP and COP are different ratings
Rated power factor	0.8	UNVERIFIED_PLACEHOLDER. Engine kW is the kVA rating times this factor
Rating basis required	PRP (prime power) per ISO 8528-1, at the site conditions	Project decision. ESP is not a basis for prolonged outages
Derating (ambient, altitude, installation)	1.0	UNVERIFIED_PLACEHOLDER. 1.00 is a placeholder
Load acceptance and transient limits	[Frequency dip, voltage dip, recovery time: project to specify]	Pending manufacturer data

5. Technical requirements

5.1 The generating set shall be rated for prime power (PRP) at the site conditions; the rating basis shall be stated on the datasheet and the nameplate.

5.2 The manufacturer shall state kW and kVA ratings, the allowed average load factor and any hours restriction for the stated basis.

5.3 The set shall accept the stated load steps, including the UPS rectifier input with its walk-in, within the specified frequency and voltage limits; the manufacturer shall provide transient response data.

5.4 The alternator shall be rated for the harmonic current of the UPS rectifier load [project to specify the harmonic data].

5.5 Minimum-load and light-load operation requirements shall be stated by the manufacturer.

5.6 Control system, paralleling, synchronising, protection, fuel system, cooling, exhaust, starting system and enclosure: [project to define].

5.7 Black-start capability and start-to-load time: [project to define].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
equipment.generator_rating	Generator PRP rating (each)	Generator set data sheet	UNVERIFIED_PLACEHOLDER
equipment.generator_rating_power_factor	Generator rating power factor	Generator set data sheet	UNVERIFIED_PLACEHOLDER
equipment.generator_esp_rating	Generator ESP rating (each)	Generator set data sheet	MISSING
equipment.generator_cop_rating	Generator COP rating (each)	Generator set data sheet	MISSING
derating.generator_ambient_factor	Generator ambient derating factor	Manufacturer derating curve for ambient conditions (generator)	UNVERIFIED_PLACEHOLDER
derating.generator_altitude_factor	Generator altitude derating factor	Manufacturer derating curve for altitude conditions (generator)	UNVERIFIED_PLACEHOLDER
derating.generator_installation_factor	Generator installation derating factor	Manufacturer derating curve for installation conditions (generator)	UNVERIFIED_PLACEHOLDER
generators.prp_max_average_load_factor_pct	PRP maximum average load factor	Generator set data sheet	MISSING
generators.operating_hours_restriction	PRP operating-hour restriction	Generator set data sheet	MISSING
generators.overload_capability	Manufacturer overload capability	Generator set data sheet	MISSING
generators.max_step_load_pct	Maximum single load step	Generator set data sheet	MISSING
generators.max_frequency_dip_pct	Maximum frequency dip allowed	Generator set data sheet	MISSING
generators.max_voltage_dip_pct	Maximum voltage dip allowed	Generator set data sheet	MISSING
generators.ups_rectifier_walk_in_s	UPS rectifier walk-in time	UPS data sheet	MISSING
generators.ups_input_current_thd_pct	UPS input current THD	UPS data sheet	MISSING
short_circuit.generator_subtransient_reactance_pct	Generator subtransient reactance x"d	Generator set data sheet	MISSING
short_circuit.generator_x_r_ratio	Generator X/R ratio	Generator set data sheet	MISSING

7. Inspection and quality requirements

7.1 Review of the set data sheet, alternator and engine data and the paralleling scheme.

7.2 Inspection of the base frame, enclosure and fuel system components.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Factory test per ISO 8528 for performance class [project to define]: load runs at agreed steps, regulation, load acceptance and transient response.

8.2 Protection and control function tests; paralleling and load sharing if applicable.

8.3 Witnessing: [owner / independent inspector to attend].

8.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Pre-start checks, fuel and coolant, protection settings.

9.2 Load bank test to rated load; transient tests with the real UPS load where possible.

9.3 Utility-failure transfer and return tests with the switchgear; fuel runtime verification against the design basis.

9.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.